Filippo Giovagnini

Mathematics PhD Candidate | Imperial College London

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LinkedIn Website GitHub
London, UK

Research Profile

Mathematics PhD candidate at Imperial College London researching deep-learning theory and stochastic analysis, with applied machine-learning experience in financial NLP and unstructured-data pipelines. Co-author of research on neural-network universality and human–AI decision making; Kaggle silver medallist in agent-driven ONNX optimisation. Python and PyTorch experience spanning mathematical research and ML implementation.

Education

Imperial College London — PhD in Mathematics – Stochastic Analysis, MFC CDT — London, UK; Dec 2023 – Present

- **Universality in Deep Neural Networks: An approach via the Lindeberg exchange principle**. With S. Kotitsas and M. Romito, 2026. Preprint.
- **Non-selection of Lagrangian trajectories in the zero-noise limit for a class of stochastic regularizations**. With L. Galeati and M. Sorella, 2026. Preprint.
- **A uniform particle approximation to the Navier–Stokes-alpha models in three dimensions with advection noise**. With D. Crisan, 2025. Preprint.
- **A uniform point vortex approximation for the solution of the two-dimensional Navier Stokes equation with transport noise**. With D. Crisan, 2024. Preprint.
- **From Information to Delegation: Mapping Human-AI Financial Decision Making**. With I. M. Bilal, Y. C. Wang, A. Raj, and others, 2026. Preprint. Study of 1.5 million ChatGPT and Gemini interactions from 6,304 users in the US and India.

University of Pisa — MSc in Mathematics – 110/110 cum laude, GPA 30/30 — Pisa, Italy; 2023

ETH Zürich — Exchange Programme – Quantitative Risk Management and Optimal Transport — Zürich, Switzerland; 2023

University of Pisa — BSc in Mathematics – 110/110 cum laude, GPA 29.85/30 — Pisa, Italy; 2021

Machine Learning Engineering

Stripe Partners — London, UK

Data Scientist, Part-time — Mar 2026 – Aug 2026

- Built a statistical NLP pipeline to extract trend indicators from social-media data and validate market-entry signals for Novo Nordisk's US strategy.
- Contributed to internal research.

Aleta Index — London, UK

Machine Learning Engineer, Part-time — Feb 2025 – Feb 2026

- Designed and built the core ranking system and platform codebase from scratch for real-time financial narrative tracking.
- Built NLP pipelines extracting topic, sentiment and framing signals from financial-news streams for structured monitoring and analysis.

Selected GitHub Projects

- **Kaggle Silver Medal – NeuroGolf 2026.** Ranked **40th (top 1.4%)** in an AI-agent and ONNX optimisation challenge. Built an iterative agent workflow to propose, test and refine compact ONNX solvers under exact-output, parameter and memory constraints. Code; competition.
- **Torchfeather – PyTorch language-model training.** Developing a hands-on project on transformer training and multi-GPU methods, including attention, RoPE and mixture-of-experts; an interactive dashboard compares training runs and generated samples. Adapted from and informed by hkproj/torchfeather. Code.

Technical Skills

Programming & frameworks: Python, PyTorch, scikit-learn, NumPy, Pandas, Polars, Git, Linux, C++.

Machine learning: Transformers, NLP, embeddings, agent workflows, ONNX optimisation, model evaluation, probabilistic modelling, feature engineering, real-time and unstructured-data pipelines.

Research methods: deep-learning theory, high-dimensional probability, stochastic calculus, SDEs/SPDEs, Monte Carlo methods, numerical analysis, statistical learning, optimal transport.

Additional: TA for MSc Stochastic Calculus for Finance at Imperial College Business School (2023–Present); Italian (native), English (fluent), Spanish (conversational).